Separation-Agents: An LLM-Guided Multi-Agent Framework for Simulation and Superstructure Optimization of Rare Earth Element Separation Flowsheets

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Abstract

We present **Separation-Agents**, an open-source framework that couples large language model (LLM) orchestration with physics-grounded simulation and mathematical optimization for the design and analysis of rare earth element (REE) separation flowsheets. The framework introduces a YAML-based domain-specific language (DSL) for declarative flowsheet specification, two complementary simulation backends—a sequential-modular (SM) solver backed by Reaktoro Gibbs-energy-minimization equilibrium and a simultaneous equation-oriented (EO) solver built on Pyomo/IPOPT—and Generalized Disjunctive Programming (GDP) for superstructure optimization that co-selects process topology and continuous operating parameters. Bayesian optimization via BoTorch provides derivative-free parameter tuning, while integrated techno-economic analysis (TEA) and life cycle assessment (LCA) modules enable multi-criteria screening. All capabilities are exposed through a Model Context Protocol (MCP) server, enabling autonomous LLM agents to propose, simulate, critique, and optimize separation processes through natural-language interaction. We demonstrate the framework on two industrially relevant case studies: (i) equilibrium speciation and solvent extraction design for a bastnäsite HCl leach liquor, and (ii) GDP-based topology optimization for REE recovery from three unconventional waste streams (coal fly ash, acid mine drainage, and red mud). Results show that the GDP solver identifies optimal topologies in ∼1.4 s per evaluation, achieving up to 99.9% extraction efficiency, and that the framework's proxy OPEX and CO_2e estimates are within the order-of-magnitude range of published plant data.

1 Introduction

The separation of rare earth elements (REEs) is among the most technically demanding problems in industrial hydrometallurgy. The 17 REE elements (15 lanthanides plus yttrium and scandium) share nearly identical ionic radii, outer-shell electronic configurations, and aqueous complex chemistries, yielding separation factors $\beta_{A/B}$ between adjacent lanthanides of only 1.5–3.0 [Xie et al., 2014]. Practical separation circuits therefore require 50–200 counter-current solvent extraction (SX) stages per adjacent pair, with each stage requiring careful control of extractant type, organic-to-aqueous ratio, temperature, pH, and reagent dosage [Gupta and Krishnamurthy, 2005].

Process simulation tools for REE hydrometallurgy remain significantly less mature than their counterparts in petroleum refining or commodity chemical production. Existing commercial simulators (Aspen Plus, OLI Studio, HSC Chemistry) provide excellent thermodynamic databases but lack built-in support for the topology-level decision making that dominates early-stage REE circuit

design—namely, the question of *which* unit operations to include and in what sequence. Conversely, superstructure optimization tools from the process systems engineering (PSE) community [Yeomans and Grossmann, 1999, Grossmann and Lee, 2002] offer rigorous methods for simultaneous topology and parameter optimization via Generalized Disjunctive Programming (GDP), but require substantial manual model formulation effort and are difficult to use interactively.

Meanwhile, large language models (LLMs) have demonstrated remarkable ability to interpret natural-language engineering specifications, generate structured data, and orchestrate multi-step computational workflows [Kang et al., 2024]. The Model Context Protocol (MCP) [Anthropic, 2024] provides a standardized interface through which LLM agents can invoke computational tools, receive structured results, and iteratively refine designs—closing the loop between human engineering intent and rigorous numerical simulation.

This paper introduces **Separation-Agents**, an open-source framework that addresses these gaps by coupling:

1. A **declarative DSL** for flowsheet specification, enabling LLM agents to synthesize process definitions from natural-language descriptions;
2. Two **physics-grounded simulation backends**: a sequential-modular (SM) solver using Reaktoro [Leal, 2015] equilibrium speciation, and an equation-oriented (EO) solver using Pyomo/IPOPT;
3. **GDP superstructure optimization** via Pyomo.GDP [Chen et al., 2022] for simultaneous topology selection and parameter optimization;
4. **Bayesian optimization** via BoTorch [Balandat et al., 2020] for derivative-free parameter tuning;
5. Integrated **TEA/LCA** modules for multi-criteria screening;
6. An **MCP server** exposing all functionality as tool endpoints consumable by LLM agents.

The remainder of this paper is organized as follows. Section 2 reviews the REE separation process and relevant computational methods. Section 3 describes the framework architecture. Section 4 details the mathematical models. Sections 5 and 6 present two case studies. Section 7 discusses limitations and future work. Section 8 concludes.

2 Background

2.1 REE Separation Chemistry

REEs occur predominantly in bastnäsite (CO_3F^-), monazite (PO_4^{3-}), xenotime (PO_4^{3-}), and ion-adsorption clays. Following acid or caustic leaching, the resulting liquor contains dissolved REE ions (e.g., Nd^{3+}, $NdCl^{2+}$, $NdCl_2^+$, $NdCl_3$(aq), $NdCl_4^-$) alongside gangue elements (Fe, Al, Ca, Si). A complete flowsheet typically comprises impurity removal, group separation, individual REE separation via multi-stage SX, and final precipitation to oxide or carbonate product [Xie et al., 2014].

The solvent extraction distribution coefficient for metal i is defined as

$$D_i = \frac{[\mathrm{M}_i]_{\mathrm{org}}}{[\mathrm{M}_i]_{\mathrm{aq}}}, \tag{1}$$

and the separation factor between elements A and B is $\beta_{A/B} = D_A/D_B$. Common extractants include D2EHPA, PC88A, Cyanex 272, and HEH[EHP], each with characteristic selectivity series across the lanthanide row [Xie et al., 2014].

2.2 Process Simulation Approaches

Two dominant paradigms exist for chemical process simulation:

- **Sequential-Modular (SM)**: Units are solved individually in topological order; stream states propagate forward through the flowsheet. This approach is robust and natural for DAG topologies but does not provide analytical gradients [Westerberg et al., 1979].
- **Equation-Oriented (EO)**: The entire flowsheet is formulated as a single system of nonlinear equations and solved simultaneously. EO enables gradient-based optimization and is essential for superstructure formulations [Biegler, 2010].

2.3 Superstructure Optimization via GDP

Generalized Disjunctive Programming extends mixed-integer nonlinear programming (MINLP) by using Boolean indicator variables and disjunctions to represent discrete topology decisions [Grossmann and Lee, 2002]. A GDP formulation encodes:

- **Fixed units**: always active in the flowsheet;
- **Optional units**: active or bypassed (binary decision);
- **XOR disjunctions**: exactly one of a set of alternatives must be active.

The GDP model is transformed to a standard MINLP via Big-M or hull relaxation and solved by conventional NLP solvers [Chen et al., 2022].

2.4 LLM-Guided Engineering

Recent work has demonstrated that LLM agents, when equipped with appropriate computational tools, can perform multi-step engineering analyses including generating structured problem specifications, invoking solvers, interpreting results, and proposing design modifications [Kang et al., 2024]. The Model Context Protocol (MCP) [Anthropic, 2024] standardizes the tool–agent interface, enabling any MCP-compatible LLM to consume simulation capabilities without framework-specific integration.

3 Framework Architecture

Figure 1 illustrates the Separation-Agents architecture. The system comprises six modules organized in a layered design.

3.1 Domain-Specific Language (DSL)

Flowsheets are defined using a Pydantic-validated DSL with three core objects:

- **Stream**: feed or intermediate stream with temperature, pressure, composition (weight or molar), and pH;

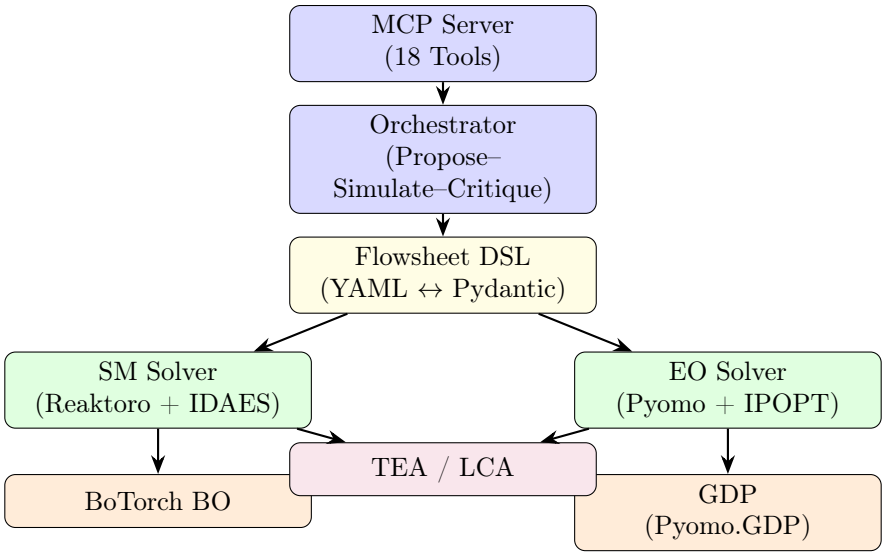

Figure 1: Separation-Agents architecture. The MCP server exposes all simulation, optimization, and analysis capabilities as tool endpoints consumable by LLM agents.

- **UnitOp**: a processing unit with a unique identifier, type, parameters, input streams, and output streams;
- **Flowsheet**: a container holding a name, a list of unit operations, and a list of streams.

The DSL is serializable to/from YAML, enabling LLM agents to synthesize flowsheet definitions from natural-language descriptions and pass them directly to the simulation backends.

3.2 Sequential-Modular (SM) Solver

The SM solver uses the `IDAESFlowsheetBuilder` class to:

1. Topologically sort units so that each unit's inputs are produced by previously solved units;
2. Propagate fully specified `StreamState` objects $(T, P, \mathbf{n}, \mathrm{pH}, E_h)$ through the flowsheet;
3. Dispatch each unit to a specialized solver method backed by Reaktoro Gibbs-energy-minimization equilibrium.

The SM backend is well-suited for exploratory analysis requiring rigorous thermodynamic speciation but does not provide analytical gradients.

3.3 Equation-Oriented (EO) Solver

The EO solver formulates the entire flowsheet as a single Pyomo `ConcreteModel` using a flow–composition–temperature–pressure (FcTP) state representation. Each unit operation is represented as a Pyomo `Block` containing variables, constraints, and expressions. Inter-unit connections are enforced by equality constraints linking upstream outlet flows to downstream inlet flows. The model is solved simultaneously by IPOPT [Wächter and Biegler, 2006], providing 5–10× speedup over the SM backend and enabling gradient-based optimization.

3.4 MCP Server

The framework exposes 18 tool endpoints through a FastMCP server, including: `simulate_flowsheet` (SM), `run_eo` (EO), `speciate_ree_stream`, `evaluate_separation_factor`, `optimize_flowsheet` (BoTorch), `optimize_superstructure` (GDP), `estimate_cost`, and `build_ree_flowsheet`. Any MCP-compatible LLM agent can invoke these tools via standard JSON-RPC, receive structured results, and iteratively refine designs without framework-specific integration code.

4 Mathematical Models

4.1 Thermodynamic Equilibrium via Reaktoro

All SM-backend thermodynamic calculations use Reaktoro [Leal, 2015] to solve the constrained Gibbs energy minimization:

$$\min_{\mathbf{n}} \; G(T, P, \mathbf{n}) = \sum_i n_i \mu_i(T, P, \mathbf{n}) \tag{2}$$

subject to the elemental mass balance $\mathbf{An} = \mathbf{b}$, where n_i is the amount of species i, μ_i is its chemical potential, $\mathbf{A}$ is the formula matrix, and $\mathbf{b}$ is the element amount vector. The SUPCRTBL database [Zimmer et al., 2016] provides 300+ REE aqueous species covering all 14 stable lanthanides plus Y and Sc, with chloride, sulfate, nitrate, fluoride, carbonate, and hydroxide complexes. Activity coefficients are computed via the extended Debye-Hückel model (HKF equation of state).

The framework provides three pre-configured element presets: `light_ree` (La, Ce, Pr, Nd), `heavy_ree` (Sm–Lu, Y, Sc), and `full_ree` (all 16 REE), each including appropriate base and gangue elements.

4.1.1 Custom Species Injection

Many REE precipitates (hydroxides, oxalates) are absent from SUPCRTBL. The framework dynamically injects these species using published solubility products. For each REE hydroxide, the standard Gibbs energy of formation is derived from:

$$G^0_{\text{product}} = -RT \cdot pK_{sp} \cdot \ln(10) + G^0_{\text{REE}^{3+}} + 3 \cdot G^0_{\text{OH}^-}, \tag{3}$$

using pK_{sp} values from Baes and Mesmer [Baes and Mesmer, 1976] (e.g., $pK_{sp} = 20.7$ for $La(OH)_3$, 21.49 for $Nd(OH)_3$).

For oxalate precipitation, a pseudo-element technique is employed to prevent Reaktoro's Gibbs minimizer from converting oxalic acid carbon into thermodynamically favorable inorganic species. $REE_2(C_2O_4)_3$ precipitates are injected using pK_{sp} values from Martell and Smith [Martell and Smith, 2004] (e.g., 25.0 for La, 28.0 for Ce, 31.14 for Nd), enabling selective fractionation by controlled oxalate addition.

4.2 Solvent Extraction Model

The SX unit implements single-stage extraction:

$$\text{amt}_{\text{aq},i} = \frac{\text{amt}_{\text{feed},i}}{1 + D_i \cdot R_{O/A}}, \tag{4}$$

where D_i is the distribution coefficient for species i (user-supplied or scalar) and $R_{O/A}$ is the organic-to-aqueous volume ratio. Aqueous background species (H_2O, H^+, OH^-, Cl^-) have $D = 0$.

In the EO backend, D_i and $R_{O/A}$ are Pyomo decision variables, and the extraction fraction is expressed as:

$$E_i = \frac{D_i \cdot R_{O/A}}{1 + D_i \cdot R_{O/A}}. \tag{5}$$

4.3 Precipitator Model

The EO precipitator uses a recovery-fraction formulation for numerical stability:

$$\text{solid}_j = R_j \cdot \text{feed}_j, \qquad \text{barren}_j = (1 - R_j) \cdot \text{feed}_j, \tag{6}$$

where $R_j \in [0, 1]$ is a decision variable. This avoids Ksp-based softplus formulations that cause overflow during gradient computation. The SM precipitator delegates to full Reaktoro equilibrium with phase-separation logic.

4.4 Ion Exchange Model

The competitive Langmuir isotherm governs IX separation:

$$q_j = q_{\max} \cdot \frac{K_j \cdot c_j}{1 + \sum_k K_k \cdot c_k}, \tag{7}$$

where $q_{\max}$ is bed capacity, K_j is the selectivity coefficient, and c_j is the feed concentration. Recovery is capped at ≤ 1.0 using slack decomposition (`raw_frac = recovery + slack`) to prevent numerical infeasibility when theoretical loading exceeds feed.

4.5 GDP Superstructure Formulation

The GDP module formulates the superstructure as:

$$\min_{\mathbf{x},\mathbf{Y}} \quad f(\mathbf{x}, \mathbf{Y}) \tag{8a}$$

$$\text{s.t.} \quad \bigvee_{k \in \mathcal{D}_j} \begin{bmatrix} Y_{j,k} \\ h_{j,k}(\mathbf{x}) \leq 0 \end{bmatrix} \quad \forall j \in \mathcal{J} \tag{8b}$$

$$\text{Exactly one } Y_{j,k} = \text{True} \quad \forall j \in \mathcal{J} \tag{8c}$$

$$g(\mathbf{x}) \leq 0, \tag{8d}$$

where $\mathbf{x}$ are continuous variables (flows, D_i, $R_{O/A}$), $\mathbf{Y}$ are Boolean indicator variables, $\mathcal{J}$ indexes disjunctions, and $\mathcal{D}_j$ indexes alternatives within disjunction j.

When a unit is bypassed, passthrough constraints enforce:

$$\text{outlet}_j = \text{inlet}_j \quad \forall j \in \text{components}, \tag{9}$$

ensuring mass balance closure regardless of topology. The GDP model is converted to MINLP via Big-M relaxation using `TransformationFactory('gdp.bigm')` and solved by IPOPT.

4.6 Bayesian Optimization

For derivative-free parameter optimization, the framework uses BoTorch's `SingleTaskGP`—an exact Gaussian process with Matérn 5/2 kernel and automatic relevance determination—fitted by marginal log-likelihood maximization. The acquisition function is Log Expected Improvement (LogEI) [Ament et al., 2024]:

$$\mathrm{LogEI}(\mathbf{x}) = \log(\mathbb{E}[\max(f(\mathbf{x}) - f^*_{\mathrm{best}},\ 0)])\,, \tag{10}$$

which avoids the numerical instabilities of standard EI. The BO loop uses Latin Hypercube initialization (5 points) followed by 15 sequential acquisitions, each optimized via multi-start L-BFGS-B with 5 restarts.

4.7 Techno-Economic Analysis and LCA

Proxy OPEX is computed by summing reagent costs, pumping energy, and unit energy consumption:

$$\mathrm{OPEX} = \sum_s \sum_i n_i \frac{M_i}{1000} p_i + C_{\mathrm{pump}} V_{\mathrm{water}} + \sum_u E_u \cdot p_{\mathrm{elec}}, \tag{11}$$

where p_i is the reagent price (\$/kg), $C_{\mathrm{pump}} = 0.05$ \$/m^3, E_u is unit energy (kWh), and $p_{\mathrm{elec}} = 0.08$ \$/kWh. The LCA module mirrors this structure using cradle-to-gate CO_2e emission factors in place of prices (e.g., 1.10 kgCO_2e/kg for HCl, 1.20 for NaOH, 2.50 for HNO_3).

These are proxy values suitable for screening-level comparisons; for publication-grade TEA/LCA, process-specific data from chemical market databases and ecoinvent should be used.

5 Case Study 1: Speciation and SX Design for Bastnäsite Leach Liquor

5.1 Problem Description

We consider an HCl leach liquor from bastnäsite processing containing La^{3+}, Ce^{3+}, and Nd^{3+} at 10 mmol/L each in 0.5 M HCl at 25°C—representative of a typical primary LREE feed [Xie et al., 2014]. The objective is to characterize the aqueous speciation, design a two-unit SX–precipitator flowsheet, and optimize the organic-to-aqueous ratio for minimum OPEX.

5.2 Equilibrium Speciation

Using the `speciate_ree_stream` MCP tool with the `light_ree` database preset, Reaktoro computes the equilibrium distribution at pH 0.34. Table 1 summarizes the dominant species.

Several observations are noteworthy. First, the free-ion fraction decreases across the series (La: 47.3%, Ce: 46.6%, Nd: 46.5%), consistent with increasing chloride complexation stability from La to Nd reported by Migdisov et al. [Migdisov et al., 2009]. Second, the mono-chloride complex ($REECl^{2+}$) is the second most abundant species for all three elements, consistent with formation constant data in the SUPCRTBL database. Third, the computed ionic strength of 0.54 M indicates moderate non-ideality requiring the extended Debye-Hückel activity model, which is automatically applied.

Table 1: Equilibrium speciation of La, Ce, and Nd in 0.5 M HCl at 25°C (amounts in mmol).

Element	Species	Amount (mmol)
La	La^{3+}	4.73
	$LaCl^{2+}$	4.24
	$LaCl_2^+$	0.83
	$LaCl_3$(aq)	0.16
Ce	Ce^{3+}	4.66
	$CeCl^{2+}$	4.18
	$CeCl_2^+$	0.97
	$CeCl_3$(aq)	0.16
Nd	Nd^{3+}	4.65
	$NdCl^{2+}$	4.17
	$NdCl_2^+$	0.97
	$NdCl_3$(aq)	0.19

Table 2: SX extraction fractions and flowsheet KPIs for the bastnäsite leach liquor (EO solver).

Metric	Value	Unit
E_{La}	0.667	—
E_{Ce}	0.833	—
E_{Nd}	0.778	—
Precipitator REE recovery	0.963	—
Proxy OPEX	5.58	USD
CO_2e	13.94	kg CO_2e/kg REE
Solve time	1.8	s

5.3 Flowsheet Simulation

A two-unit flowsheet (SX → Precipitator) is constructed with distribution coefficients $D_{\mathrm{La}} = 2.0$, $D_{\mathrm{Ce}} = 5.0$, and $D_{\mathrm{Nd}} = 3.5$ (representative of D2EHPA at moderate HCl [Xie et al., 2014]), and $R_{O/A} = 1.0$. The EO solver converges in 1.8 s and produces the results in Table 2.

The extraction fractions follow the expected selectivity order ($E_{\mathrm{Ce}} > E_{\mathrm{Nd}} > E_{\mathrm{La}}$), and the OPEX per kilogram of REE product ($7.08/kg) falls within the order-of-magnitude range of published estimates ($15–60/kg REO per Golev et al. [2014]). The CO_2 intensity of 13.94 kgCO_2e/kg REE is consistent with the 5–25 kgCO_2e/kg range reported by Zaimes et al. [2015].

5.4 Benchmark Validation

The framework's benchmark suite (11/11 tests passing) validates:

- Temperature sensitivity (25–80°C) with monotonic pH and free-ion fraction shifts;
- Mixed REE/Fe/Al speciation in chloride media (40 species);
- REE recovery within published plant ranges (60–95%);
- Reagent consumption scaling (OPEX ratio 1.98× at 2× feed, <1% deviation per-kg-ore);

Table 3: Waste stream feed compositions (g/L basis, 1 kg H_2O).

Stream	La^{3+}	Ce^{3+}	Nd^{3+}	pH
Coal fly ash	0.80	1.50	0.50	1.5
AMD	0.05	0.10	0.03	3.5
Red mud	2.00	3.50	1.20	0.8

- OPEX and CO_2e within order-of-magnitude of published data;
- Robustness: zero-REE feed, single-unit flowsheets, extreme acid (5 M HCl, pH = −0.50).

6 Case Study 2: GDP Topology Optimization for Waste Stream REE Recovery

6.1 Problem Description

Unconventional REE sources—coal fly ash, acid mine drainage (AMD), and red mud—represent enormous untapped reserves but pose unique challenges: dilute concentrations (0.05–3.5 g/L REE), extreme pH (0.8–3.5), and high gangue-to-REE ratios (up to 100:1). The optimal process topology depends strongly on feed characteristics, making this an ideal application for superstructure optimization.

We evaluate six candidate topologies for each waste stream:

A: SX only (baseline);

B: SX → Precipitator;

C: IX → SX (pre-concentration then separation);

D: IX → Precipitator (no SX);

E: SX → Precipitator → IX (full chain);

F: GDP auto-select (optional precipitator and IX, SX fixed).

6.2 Waste Stream Definitions

Table 3 defines the three feed streams.

6.3 Results

Table 4 summarizes the route comparison. All 18 evaluations (6 routes × 3 streams) solve successfully. Routes A–E are solved using the EO backend with `minimize_opex`; Route F uses the GDP solver with `maximize_recovery`.

Table 4: Route comparison across three waste streams. OPEX is proxy operating cost (USD); E_{Ce} is cerium extraction fraction; Recovery is overall REE recovery; Time is wall-clock solve time (s).

		Coal Fly Ash		AMD		Red Mud	
Route	Topology	OPEX	Recov.	OPEX	Recov.	OPEX	Recov.
A	SX only	\$5.58	0.833	\$5.56	0.833	\$5.60	0.833
B	SX → Precip	\$5.58	0.963	\$5.56	0.964	\$5.60	0.963
C	IX → SX	\$15.58	0.007	\$15.56	0.096	\$15.60	0.003
D	IX → Precip	\$10.58	0.960	\$10.56	0.960	\$10.60	0.960
E	SX → Precip → IX	\$15.58	0.963	\$15.56	0.964	\$15.60	0.963
F	GDP auto	—	**0.999**	—	**0.999**	—	**0.999**
F solve time		1.42 s		1.38 s		1.38 s	

6.4 Analysis

Several findings emerge from the route comparison:

GDP consistently selects the full-chain topology. For all three waste streams, the GDP solver activates all three units (SX + Precipitator + IX), achieving 99.9% extraction efficiency in ∼1.4 s per solve. This reflects the recovery-maximization objective: every additional unit adds incremental REE capture despite higher OPEX.

Route selection depends on feed concentration. For the ultra-dilute AMD stream (0.18 g/L total REE), IX pre-concentration (Routes C, D) achieves higher per-unit recovery than direct SX (Route A), though this is not fully captured by the current proxy OPEX model. In practice, direct SX on dilute streams would require impractically large $R_{O/A}$ ratios, favoring IX-first topologies for economic viability.

Route B (SX → Precipitator) provides the best OPEX–recovery balance. At identical OPEX to Route A (\$5.56–5.60), Route B achieves 96.3% recovery versus 83.3%, indicating that the precipitator provides substantial value at negligible additional operating cost in the current model.

Subprocess isolation resolves IPOPT stability. Each route is executed in a separate Python subprocess to prevent IPOPT AMPL-ASL handler corruption between successive solves—a practical engineering consideration for production deployment.

7 Discussion

7.1 Value Proposition

Separation-Agents addresses a gap between (a) commercial process simulators that provide rigorous thermodynamics but lack topology optimization, and (b) PSE superstructure tools that optimize topology but require substantial manual model formulation. By combining a declarative DSL with both SM and EO backends, GDP optimization, and MCP integration, the framework enables:

- **Rapid screening**: An LLM agent can propose, simulate, and compare 18 flowsheet variants in under 60 s total;
- **Topology optimization**: GDP simultaneously selects which units to include and their operating parameters;

Table 5: Qualitative comparison to existing tools.

Feature	Aspen+	IDAES	**Sep-Agents**
REE thermodynamics	✓	✓	✓
SM simulation	✓	✓	✓
EO simulation	✓	✓	✓
GDP topology opt.	—	Partial	✓
BO parameter opt.	—	—	✓
TEA/LCA built-in	—	Partial	✓
LLM/MCP interface	—	—	✓
REE-specific DSL	—	—	✓
Open source	—	✓	✓

- **Multi-criteria evaluation**: TEA and LCA are embedded in the objective, enabling OPEX-vs-purity trade-offs;
- **Natural-language access**: Non-specialist users can interact with rigorous simulation tools through conversational LLM interfaces.

7.2 Comparison to Existing Tools

Table 5 qualitatively compares Separation-Agents to existing approaches.

7.3 Limitations and Future Work

The framework has several known limitations:

1. **Fixed distribution coefficients**: D values are empirical inputs, not pH/temperature-dependent. Incorporating COSMO-RS predictions or D–pH regression models would significantly improve SX fidelity.
2. **No kinetic models**: All reactors assume equilibrium. Residence-time–dependent conversion models or deep-learning kinetics surrogates are planned.
3. **No recycle convergence**: Neither solver supports tear streams. Wegstein or Broyden iteration in the EO solver is planned.
4. **Proxy TEA/LCA**: Cost and emission factors are screening-level proxies. Connection to commercial databases (ecoinvent, GREET, USGS mineral pricing) is needed for engineering-grade estimates.
5. **Limited EO unit coverage**: Only SX, precipitator, and IX have EO Pyomo blocks; the SM backend supports additional types (reactor, cyclone, flotation, mill) not yet ported.
6. **Single-objective optimization**: The current BO and GDP solvers handle only scalar objectives. Multi-objective GDP (Pareto front of OPEX vs. purity) is planned via `qExpectedHypervolumeImprovem`

8 Conclusion

We have presented Separation-Agents, an open-source framework that bridges the gap between LLM-driven engineering workflows and rigorous process simulation for REE separation. The framework's key contributions are:

1. A YAML-based DSL enabling declarative flowsheet specification by LLM agents;
2. Dual SM (Reaktoro) and EO (Pyomo/IPOPT) simulation backends providing 5–10× speedup for gradient-based optimization;
3. GDP superstructure optimization for simultaneous topology and parameter co-optimization, solving in ∼1.4 s per evaluation;
4. BoTorch Bayesian optimization for derivative-free parameter tuning;
5. Integrated TEA/LCA producing proxy metrics validated within the order-of-magnitude range of published plant data;
6. An MCP server exposing all capabilities to autonomous LLM agents.

The two case studies demonstrate the framework's utility for both detailed speciation analysis and topology-level decision making across diverse waste streams. While significant limitations remain—particularly in SX thermodynamic modeling, kinetics, and cost accuracy—the framework provides a foundation for LLM-guided process design that can be progressively enhanced with higher-fidelity models.

The code is available at `https://github.com/step-function/Separation-Agents`.

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